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Equilibrium Drift Restriction: A Control Strategy for Reducing Steady-State Error Under System Inconsistency

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Abstract

The inconsistency of system parameters inevitably emerges due to reasons such as modeling imprecision, manufacturing error, and aging process. Due to the inconsistency between nominal models and real-world conditions, controllers designed accordingly frequently fail to maintain performance guarantees during physical deployment. This phenomenon exemplifies the open sim-to-real gap problem. To address this limitation, we develop an equilibrium drift restriction strategy (EDR) to reduce the steady-state error due to the system inconsistency. We first present an example to show the reason why some existing controllers cannot counteract the system inconsistency when the equilibrium is not at the origin. Then, a control strategy is proposed by using the EDR method to reduce the induced steady-state error. Both intuitive interpretation and theoretical analysis demonstrate how EDR reduces steady-state deviations. Simulation results of a common pendulum system are provided to demonstrate that the restriction mitigates the impact of parameter inconsistency. A comparison with the popular Q-learning method is also presented. The results show that the EDR method can serve as a simple but effective tool to improve the steady-state performance of existing controllers. This paper offers a fresh perspective for exploring the control functions with specific properties in the realm of related controller research.

Keywords: system inconsistency; regulation control; equilibrium drift

1. Introduction

During batch production and operational aging, the system parameters inevitably drift from their nominal values. Consequently, this system parameter inconsistency arises as a persistent challenge [1]. In modern power systems, the widespread integration of power converters has led to a progressive reduction in system inertia and damping capacity. While these devices inherently exhibit parameter variability across units, their operational characteristics necessitate advanced control strategies for effective frequency regulation [2]. Additionally, factors such as model uncertainties and imprecise representations further exacerbate system inconsistency. These discrepancies may originate from mismatches between theoretical models and actual system behaviors or from inherent differences among heterogeneous plant components. Such parameter-induced inconsistency ultimately undermines the overall stability and performance of the power grid.

Such control-related challenges are conventionally analyzed and mitigated through the theoretical frameworks of robust control. Many methods, including the small-gain approach [3], Lyapunov method [4], pole placement [5] have long been studied. Since system performance is most directly determined by the eigenvalues of the parameter matrix (or system poles), the structural properties such as eigenvalue sensitivities [6–8]

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have also been extensively studied. Researchers in [9] developed a robust model predictive control strategy tailored for systems subjected to additive disturbances and model uncertainties. Building on similar system structures, reference [10] introduced a robust recursive linear quadratic regulator framework to address discrete-time linear systems with unknown yet bounded state delays. However, the challenge of incomplete or inaccurate modeling—whether due to unmodeled dynamics or partial knowledge of plant characteristics—remains an unresolved issue in control theory. In response, recent research has increasingly focused on model-free approaches, such as policy-based optimization techniques (e.g., policy gradient methods) and value-based reinforcement learning algorithms (e.g., Q-learning) [11–13]. Reference [13] further advanced this domain by proposing an off-policy model-free framework grounded in optimization principles for linear time-invariant systems. For heterogeneous multi-agent systems, distributed data-driven control strategies leveraging reinforcement learning have been suggested to achieve robust synchronization without requiring precise knowledge of system dynamics [14]. Additionally, researchers in [15] proposed a distributed Q-learning-based unit commitment method to address infinite-horizon scheduling problems under potential disturbances. Meanwhile, deep reinforcement learning approaches have been adapted for power systems to enhance resilience against dynamic environmental changes [16]. Despite these advancements, achieving satisfactory performance typically demands substantial training data and intricate network architecture design [17].

Given the data-intensive nature of data-driven methods and the persistent sim-to-real gap [18], model-based approaches provide an efficient and computationally lighter alternative. This study introduces the EDR strategy as a novel framework to address system parameter inconsistency. Unlike data-driven methods, EDR does not rely on extensive training data. Instead, it offers an analytical approach to design control functions with predefined stability and performance properties, thereby expanding the toolkit for controller development. To clarify its position relative to existing paradigms: classical robust control often guarantees worst-case performance at the cost of conservatism and computational complexity; reinforcement learning adapts to uncertainty through data but suffers from high sample complexity and limited extrapolation. In contrast, EDR directly targets the root cause of steady-state error and provides a lightweight modification to existing controllers without retraining or worst-case analysis. The main contributions are summarized as follows:

1. Analysis is conducted to show the steady-state deviations under parameter inconsistency in classical control architectures, explaining conventional methods' limitations in practical applications.
2. Geometric intuition on the principle of the proposed EDR strategy is presented. Theoretical results also provide rigorous guarantees for deviation bounds and convergence properties under both nominal and perturbed conditions.
3. Comprehensive validations are also provided to show the ability of preserving nominal tracking performance, enhancing robustness against parameter variations, superior adaptability compared to Q-learning control, and sensitivity reduction through parameter adjustment.

The remainder of this paper is organized as follows: Section 2 presents a case study based on a linear time-invariant (LTI) system to highlight the shortcomings of conventional control methodologies. Section 3 presents the geometric intuition and theoretical analysis of the proposed EDR strategy. Then, comprehensive validations of the proposed strategy are conducted on a pendulum system in Section 4. Finally, the conclusions of this paper are drawn in Section 5.

2. Classical Methods and Their Limitations

This section examines classical reference tracking methods and their inherent limitations as a representative example. Consider a representative LTI system as follows:

$$\dot{x} = Ax + Bu, \tag{1}$$
$$y = Cx + Du, \tag{2}$$

where $x \in \mathbb{R}^n$ is the state–space variable. $u \in \mathbb{R}$ is the control input. $y \in \mathbb{R}$ is the selected output variable of interest. A, B, C and D are the system matrices of appropriate dimension.

If the control input function u is equal to zero, one has the autonomous system calculated as follows:

$$\dot{x} = Ax, \tag{3}$$
$$y = Cx. \tag{4}$$

Assumingan initial condition $x(0) = x_0$, the state trajectory is given by $\dot{x}(t) = p_i e^{p_i t}$ $x_0 = Ax = Ae^{p_i t}x_0$. Specifically, stability is characterized by the eigenvalues p_i of A, which satisfy $\det(p_i I - A) = 0$.

The above analysis of the LTI system is necessary and useful since many problems can be transformed into this form eventually. The following presents the usual design process of a stabilized controller. If the system is not stable, state feedback controllers are usually introduced such that the modified system can be stabilized. Assume that $u = -Kx$, then, the system (1) and (2) are modified as follows:

$$\dot{x} = (A - BK)x, \tag{5}$$
$$y = (C - DK)x. \tag{6}$$

Then, the stability of the controlled system is characterized by $\det[p_i I - (A - BK)] = 0$. In the usual sequential process, the controller can be designed by picking the gain K such that the roots lie in desirable locations. Up to this point, the resulting controller with gain K still retains robustness to the system parameters. The reason is that one can pick gain and make eigenvalues p_i negative enough.

To drive the system state to some new setpoint, the error system is a popular tool such that the related controller design is then transformed into a previous stability problem around equilibrium. However, it is in this procedure when the robustness to system parameters is weakened. The reasons are explained in the remainder of this section.

Suppose the new state at steady-state is x_s (y_s for output), then one has the following:

$$0 = Ax_s + Bu_s, \tag{7}$$
$$y_s = Cx_s + Du_s. \tag{8}$$

Thus, the error system is as follows:

$$\dot{e} = Ae + B(u - u_s), \tag{9}$$
$$y - y_s = Ce + D(u - u_s). \tag{10}$$

where $e = x - x_s$ represents the state error. The control law is defined as follows:

$$u - u_s = -Ke. \tag{11}$$

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One can then select some proper gain vector K such that the error dynamic is stable around equilibrium 0 as in the previous situation. The remaining problem is to solve for the state x_s, output y_s, and control u_s at steady-state. Since the y_s is the setpoint that one expects the system to stay at, thus it can be treated as known, then the following is calculated:

$$\begin{bmatrix} 0 \\ y_s \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} x_s \\ u_s \end{bmatrix}. \tag{12}$$

Assuming the matrix $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ is invertible, the steady-state values can be obtained as follows:

$$\begin{bmatrix} x_s \\ u_s \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ y_s \end{bmatrix}. \tag{13}$$

Equation (13) computes the steady-state state–input pair (x_s, u_s) required to maintain output y_s. Finally, the full controller with reference input is derived as $u = u_s - K(x - x_s)$ with x_s and u_s calculated by using (13).

Critically, while the error dynamics $\dot{e} = (A - BK)e$ are robustly stabilized, the steady-state target (x_s, u_s) depends explicitly on all system parameters via (13). Therefore, any inconsistency between the nominal model and the true plant directly corrupts the reference trajectory, leading to an irreducible steady-state error. This amplification of parameter inconsistency through equilibrium calculation is a key limitation of conventional tracking architectures. Therefore, the invariant reference y_s cannot achieve robustness to inconsistency due to the dependents in (13).

3. Proposed Equilibrium Drift Restriction Strategy

This section presents the design and analysis of the EDR strategy. Suppose the practical system matrix, due to manufacturing errors or the aging process, is $\hat{A}$ with some distribution, while its nominal value is A. In practice, however, the computed steady-state values x_s and u_s may no longer satisfy the following:

$$\begin{bmatrix} 0 \\ y_s \end{bmatrix} = \begin{bmatrix} \hat{A} & B \\ C & D \end{bmatrix} \begin{bmatrix} x_s \\ u_s \end{bmatrix}, \tag{14}$$

due to parameter inconsistency. For convenience, we focus on the case when C and D are identity and zero matrices. The other parameters B, C, and D could also be investigated similarly.

Analysis of (14) reveals a critical insight that if x_s and u_s can be designed with the impact of parameters mitigated, the resulting controller will be able to counteract the parameters' inconsistency. The inconsistency affects equilibrium through the first equation of (14) as follows:

$$Bu_s = -\hat{A}x_s. \tag{15}$$

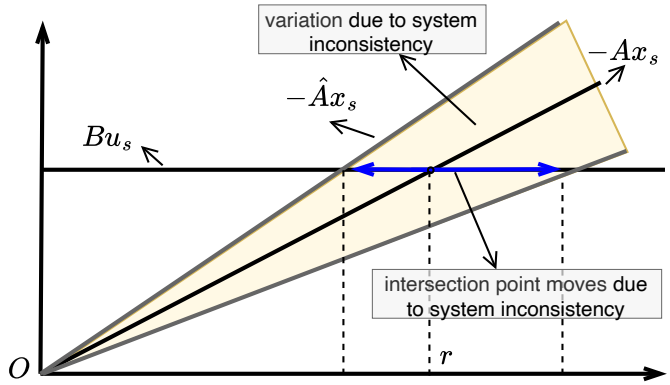
Geometrically, this represents the intersection of two functions: $f(x_s) = -\hat{A}x_s$ and $g(x_s) = Bu_s$. For an intuitive understanding, let us take one dimensional case as an example, whose intersection point is shown in the following Figure 1a.

As shown in Figure 1a, the intersection point shifts with changes in the slope of $f(x_s) = -\hat{A}x_s$. With the controller (11) in Section 2, such a shift is unavoidable. Take a further step along the intersection point of view. However, if one can select the function $g(x_s)$ as some variant function as shown in Figure 1b other than $Bu_s(x_s)$, then the x_s of the

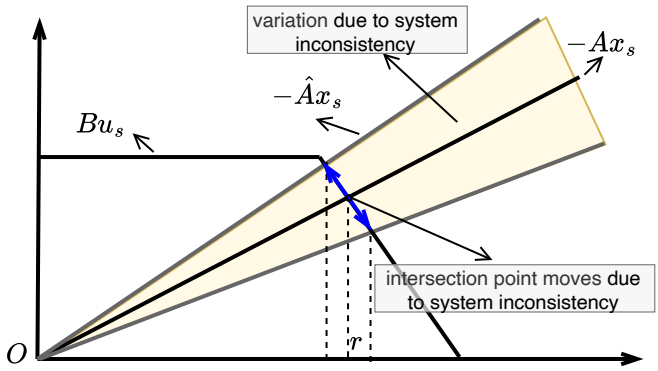
intersection point could be more invariant with respect to $\hat{A}$. Thus, the impact of parameter inconsistency can be largely reduced. With this observation, the simplest choice of such a variant function is a linear correction term, yielding the following:

$$f(x_s) = Bu_s - \Gamma(x_s - r), \tag{16}$$

where $r \in \mathbb{R}^n$ is the desired equilibrium, and $\Gamma \in \mathbb{R}^{n\times n}$ is a design matrix that governs the strength and direction of the corrective feedback. In practice, one can also select a special form of Γ, such as $diag\{\alpha_1, ..., \alpha_n\}$, as a candidate weight matrix. Larger eigenvalues of Γ increase the "pull" toward the reference r. Other candidate functions could also be possible, such as logistic function, hyperbolic tangent, and smooth-step function. Although nonlinear alternatives may also be possible, the linear form preserves analytical tractability and integrates seamlessly with LTI theory.



(**a**) Constant reference



(**b**) Variant reference

Figure 1. Geometric interpretation of system equilibrium under constant versus variant reference strategies.

Then, the new equilibrium equation becomes the following:

$$Bu_s = -\hat{A}x_s + \Gamma(x_s - r). \tag{17}$$

Comparing (17) with (15) and (13), one obtains that the modification approach adds a term that is variable with respect to the state. This constitutes the core mechanism of EDR. In the following, we differentiate between the case when there is no system inconsistency and then proceed to the case when system inconsistency exist. Instead of studying how the system

converges to its equilibrium, the following two theorems present how the equilibrium moves due to the above-proposed EDR strategy.

Theorem 1. *Consider the following LTI system:*

$$\dot{x} = \hat{A}x + Bu,$$

and suppose that the control input u_s is designed according to the proposed law

$$Bu = -\hat{A}x + \Gamma(x - r),$$

where $r \in \mathbb{R}^n$ is a constant reference vector, and $\Gamma \in \mathbb{R}^{n\times n}$ is a constant weight matrix. If Γ is Hurwitz and there is no system inconsistency, then the equilibrium point $x_s = r$ is globally exponentially attractive. Moreover, the rate of convergence is governed by the eigenvalues of Γ.

Proof. Since it has been assumed no system inconsistency in this case, one can easily obtain the following:

$$\dot{x} = \Gamma(x - r) \tag{18}$$

by substituting the proposed law into the system dynamic. Because Γ is Hurwitz by assumption, all eigenvalues of Γ lie strictly in the open left-half complex plane. It follows that the equilibrium $x_s = r$ is globally exponentially stable. The convergence rate is determined by the eigenvalues of the weight matrix Γ. Thus, this readily proves the result. □

Theorem 1 demonstrates that the proposed EDR strategy preserves stability provided that the weight matrix Γ is negative definite and there is no system inconsistency. However, when system inconsistency is present (i.e., $\hat{A} \neq A$), the equilibrium may no longer be exactly attainable. In this case, the following result characterizes the resulting steady-state deviation in terms of the mismatch $\hat{A} - A$ and the choice of Γ.

Theorem 2. *Consider the following LTI system:*

$$\dot{x} = \hat{A}x + Bu,$$

where $\hat{A} \in \mathbb{R}^{n\times n}$ denotes the actual system matrix and $A \in \mathbb{R}^{n\times n}$ its nominal counterpart. Assume that the control input is implemented according to the following proposed law:

$$Bu = -Ax + \Gamma(x - r),$$

with a constant reference vector $r \in \mathbb{R}^n$, and where $\Gamma \in \mathbb{R}^{n\times n}$ is a nonsingular matrix. Then, the steady-state deviation is bounded by the following:

$$||x_s - r|| \leq ||\Gamma^{-1}(\hat{A} - A)|| \cdot ||x_s||. \tag{19}$$

Moreover, if the matrix $\Gamma + \hat{A} - A$ is Hurwitz, then the resulting equilibrium x_s is globally exponentially attractive, and the rate of convergence is governed by the eigenvalues of $\Gamma + \hat{A} - A$.

Proof. Since there is system inconsistency, one cannot design the controller with the real $\hat{A}$, but the nominal A. Then, one has the following:

$$Bu = -Ax + \Gamma(x - r). \tag{20}$$

Substituting it into the system dynamic, one obtains the following:

$$\dot{x} = (\hat{A} - A)x + \Gamma(x - r), \tag{21}$$

and steady-state

$$0 = (\hat{A} - A)x_s + \Gamma(x_s - r), \tag{22}$$

Taking normalization at both sides, one obtains the bound (19). It follows from (21) that the convergence of x to x_s is global exponential if $\Gamma + \hat{A} - A$ is Hurwitz. The convergence rate is also determined accordingly. □

From (19) we know that when the new equilibrium x_s is not zero, the following holds:

$$\frac{||x_s - r||}{||x_s||} \leq ||\Gamma^{-1}(\hat{A} - A)||.$$

Thus, larger absolute eigenvalues of Γ and smaller system inconsistency contribute to smaller steady-state deviation. One can also select Γ to be "negative" enough to improve convergence speed, which is coincident with existing knowledge. However, if the new equilibrium is the origin, i.e., $x_s = 0$, then neither system inconsistency nor Γ matters anymore, which also confirms the analysis in the previous section. In the above results, modification (17) is used directly as control input to study the drift of equilibrium caused by system inconsistency. It is also applicable to combine the modification (17) with (11) to obtain a new controller as demonstrated in the following section.

Remark 1. *It should also be mentioned that the geometric interpretation has relative wider applications not limited to the above discussions on the continuous-time LTI system. Consider the following discrete-time LTI system:* $x[k+1] = \hat{A}x[k] + Bu[k]$, *where the equilibrium satisfies* $Bu_s = (I - \hat{A})x_s$. *By using the similar geometric interpretation of the EDR framework, the analogous law becomes as follows:* $Bu[k] = (I - A)x[k] + \Gamma(x[k] - r)$, *yielding closed-loop dynamics* $x[k+1] = (I + \hat{A} - A)x[k] + \Gamma(x[k] - r)$. *By choosing* Γ *appropriately, the system can also be stabilized with steady-state deviation, written as follows:* $||x_s - r|| \leq ||\Gamma^{-1}(\hat{A} - A)|| \cdot ||x_s||$.

Remark 2. *Among all affine correction terms of the form* $g(x) = Mx + c$, $M \in \mathbb{R}^{n\times n}$, $c \in \mathbb{R}^n$, *the specific choice* $g(x) = \Gamma(x - r)$ *offers three advantages: (1) In the absence of system inconsistency, this correction preserves the desired equilibrium* $x_s = r$, *as formally established in Theorem 1. (2) It results in linear closed-loop error dynamics* $\dot{e} = \Gamma e$, *which are analytically tractable and guarantee exponential stability provided that* Γ *is Hurwitz. (3) While nonlinear correction functions could be conceived, they generally compromise the linearity of the closed-loop system and complicate stability analysis. Although such nonlinear alternatives might offer performance benefits in certain scenarios, their efficacy in the LTI system remains an open question and warrants further investigation.*

4. Validation Through Simulation

In this section, the effectiveness of the proposed EDR strategy is demonstrated via two representative examples.

Consider an undampened pendulum system, written as follows:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\hat{\omega}_0^2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u,$$

and

$$y = x_1,$$

where x_1 and x_2 denote the angular displacement and angular velocity. $\hat{\omega}_0$ denotes the intrinsic frequency.

A full-state feedback controller with constant reference input r can be given as follows [19]:

$$u = -Kx + (N_u + KN_x)r, \tag{23}$$

where $N_x \in R^{2\times 1}$ and $N_u \in R$ satisfy

$$\begin{bmatrix} 0 & 1 & 0 \\ -\omega_0^2 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} N_x \\ N_u \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T.$$

Suppose the gain K is chosen as $[3\omega_0^2\ 4\omega_0]$ with $\omega_0 = 1$. Then, a controller with constant reference input is obtained as follows:

$$u = -3x_1 - 4x_2 + 4r. \tag{24}$$

Applying the proposed EDR strategy on (24), a new controller with variant reference input can be obtained as follows:

$$u = -(3 + \alpha^2)x_1 - 4x_2 + (4 + \alpha^2)r, \tag{25}$$

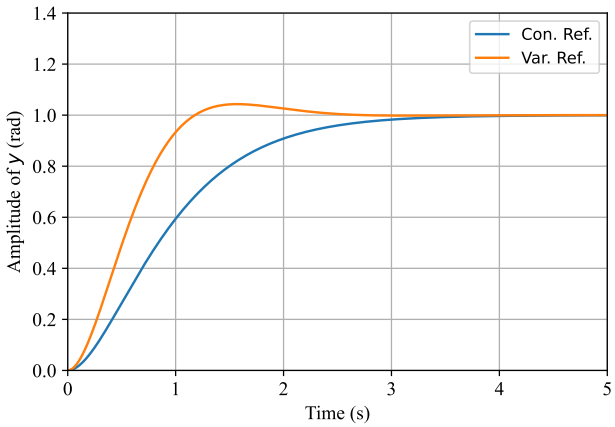
where α is a parameter determining the skew rate similarly in Figure 1b. Though the last term $(4 + \alpha^2)r$ seems constant w.r.t. output y (state x_1), the whole modification is indeed variant to the output y (state x_1). Specifically, the term $4r - \alpha^2(x_1 - r)$ before rearranging in (25) response to the x_1 variation due to system inconsistency. It can also be observed that (24) can be seen as a special case of (25) when $\alpha = 0$.

4.1. Nominal System Performance Validation

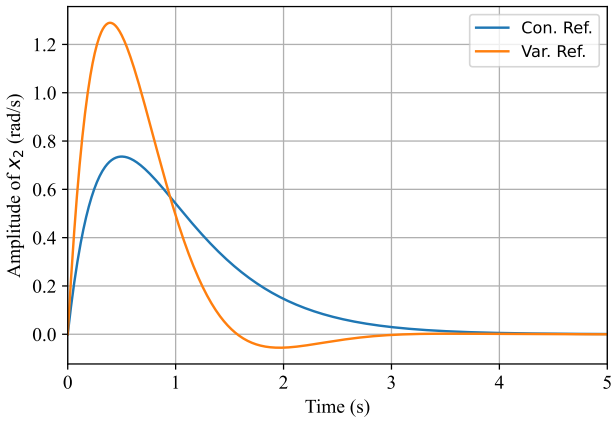
Assume the pendulum system has no inconsistency between the design stage and the test stage. This validation under nominal conditions confirms baseline performance. Let the setpoint be $r = 1$. The parameter α is set as 2 for the controller with variant reference. The step responses of the system under both constant and variant reference are plotted in Figure 2a,b. It can be observed that both of the two controllers can regulate the output y (state x_1) to the value $r = 1$. The state x_2 converges to zero at steady-state correctly. The closed-loop system under (25) can be theoretically verified to be stable, i.e., converges to its equilibrium. This simulation case shows that the equilibrium does not change under the two controllers when there is no inconsistency. The controller with variant reference also responds faster due to the existence of the additional term.

4.2. Robustness Evaluation Under Parameter Variations

Robustness evaluation considers intrinsic frequency $\hat{\omega}_0$ variations $\hat{\omega}_0 \in [0.5, 1.5]$. Set $\omega_0 = 1$, $r = 1$, and $\alpha = 2$. The step responses of the system under controllers (24) and (25) are shown in Figure 3a,b. Figure 2a shows that the output y and state x_1 reach the constant reference input $r = 1$ as expected, since both of the controllers are designed exactly based on $\omega_0 = 1$. If the system inconsistency occurs, as shown in Figure 3a, the output y and state x_1 will not be controlled to reference input $r = 1$ anymore. This result verifies previous analysis that the equilibrium drifts away from $[1, 0]^T$ due to the system inconsistency. On the other hand, one can also observe from Figure 3b that the system inconsistency does not affect the final value of state x_2, since the variation of ω_0 does not move the x_2 of the equilibrium.



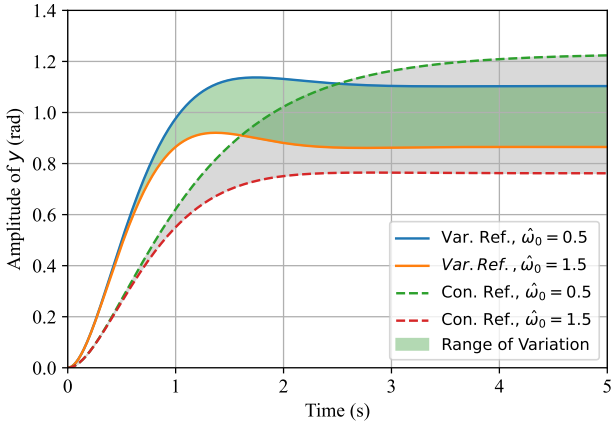
(**a**) Amplitude of y



(**b**) Amplitute of x_2

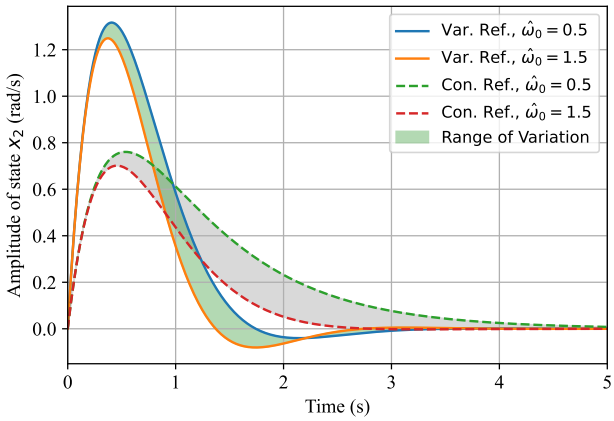
Figure 2. Comparison of the pendulum system response in the absence of system inconsistency between constant and variant reference input.

Comparing the results under the two controllers, one can observe that the variation range of steady-state y (state x_1) under the modification is narrower around the target value, showing the benefit of the proposed strategy.



(**a**) Amplitude of y

Figure 3. *Cont.*



(**b**) Amplitude of x_2

Figure 3. Comparison of system response under different system inconsistency between constant and variant reference input.

4.3. Comparison Study with Q-Learning Control

Comparative assessment employs Q-learning with cost function:

$$R(x,u) = (x_1 - 1)^2 + 0.01x_2^2, \tag{26}$$

where u is not included in this cost since it should not be zero for equilibrium $[1,0]^T$. The discount factor and learning rate of Q-learning are set as 0.99 and 0.1. The state space is set as $(-\pi/2, \pi/2)$, which is discretized uniformly. The action space is set as $\{-1,0,1\}$. The Q-function is randomly initialized uniformly in $[0,1]$. The update of the Q-function is written as follows:

$$Q(x,u) \leftarrow Q(x,u) + \alpha_l\big(R(x,u) + \gamma \min_{u'} Q(x',u') - Q(x,u)\big). \tag{27}$$

The optimal policy is obtained as follows:

$$u(x) = \arg\min_{u} Q(x,u). \tag{28}$$

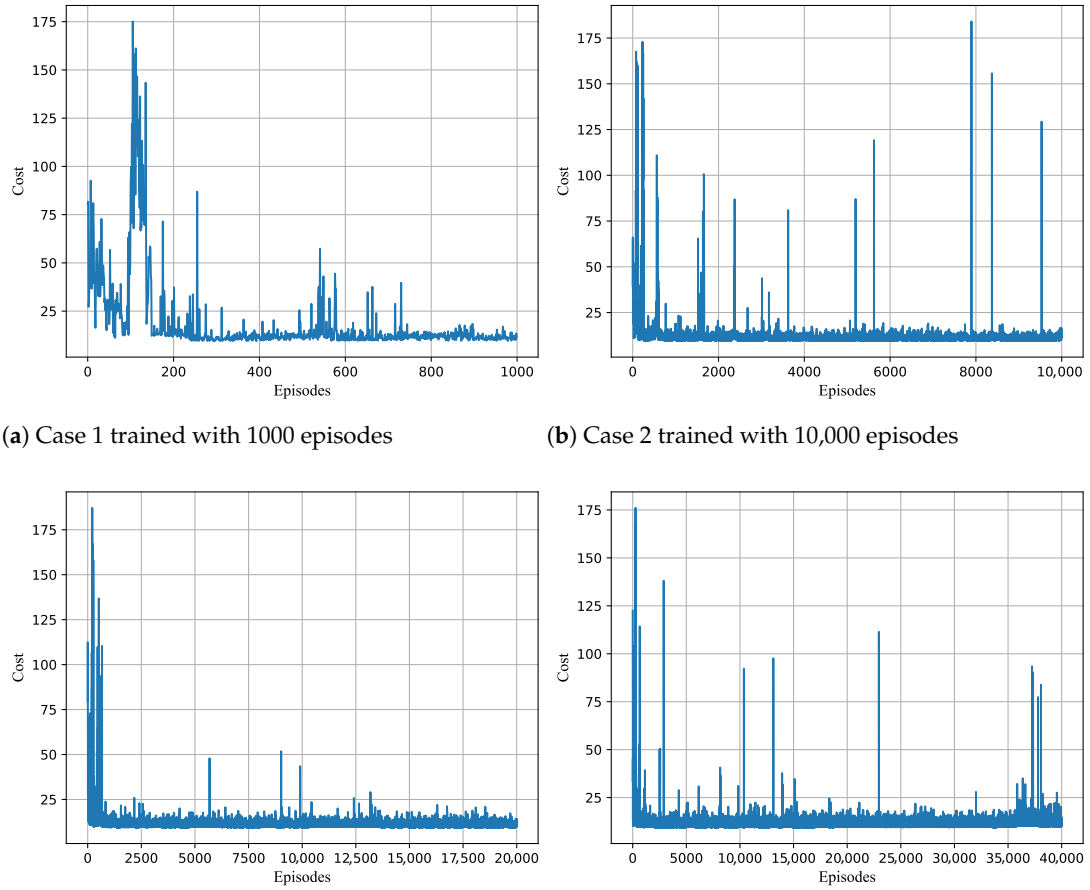
The above cost function definition is proper since the target equilibrium $[1,0]$ is its minimum value. The discount factor is set as $\gamma = 0.99$. The learning rate is set as $\alpha_l = 0.1$. The initial state for the pendulum is $x = [0,0]^T$. The state space is set as $(-\pi/2, \pi/2)$, which is then discretized. The action space is set as $\{-1,0,1\}$. The initial value of Q is set randomly and uniformly in $[0,1]$. The simulation time for each episode is 5s. The training episodes are 1000, 10,000, 20,000, and 40,000, respectively. The cost-per-episode results are in Figure 4 as follows.

Generally, larger trained episodes tend to give better-trained policy, but not necessarily. From the above cost-per-episode results, it can be seen that the trained result with 20,000 episodes is better than with 1000 and 10,000 episodes. However, the trained result with 40,000 episodes has better initial costs but worse final costs.

To choose the best one for comparison with the proposed strategy, we test the four trained optimal policies on the pendulum system with inherited $\hat{\omega}_0 = 1$, $\hat{\omega}_0 = 0.5$, and $\hat{\omega}_0 = 1.5$. The test results are shown in Figure 5 as follows.

We focus on the blue line where the policy is trained and tested on the same system. From the above results, one can also observe that the training with 20,000 episodes performs better than the others when there are system inconsistencies such as $\hat{\omega}_0 = 0.5$ and $\hat{\omega}_0 = 1.5$. For the first case with 1000 trained episodes, though the system is both trained and tested

with $\hat{\omega}_0 = 1$, the trained optimal policy can only regulate the state x_1 between 0.7 and 1.2. Thus, this number of episodes is not enough to train the policy. For the second case with 10,000 episodes, one can observe that the trained policy can regulate x_1 between 1.0 and 1.1. A larger number of training episodes should be applicable. For the third case with 20,000 episodes and the fourth case with 40,000 episodes, the trained policy can both regulate x_1 around 1 when $\hat{\omega}_0 = 1$ and $\hat{\omega}_0 = 1.5$. However, it is not guaranteed that larger trained episodes must render better policy.



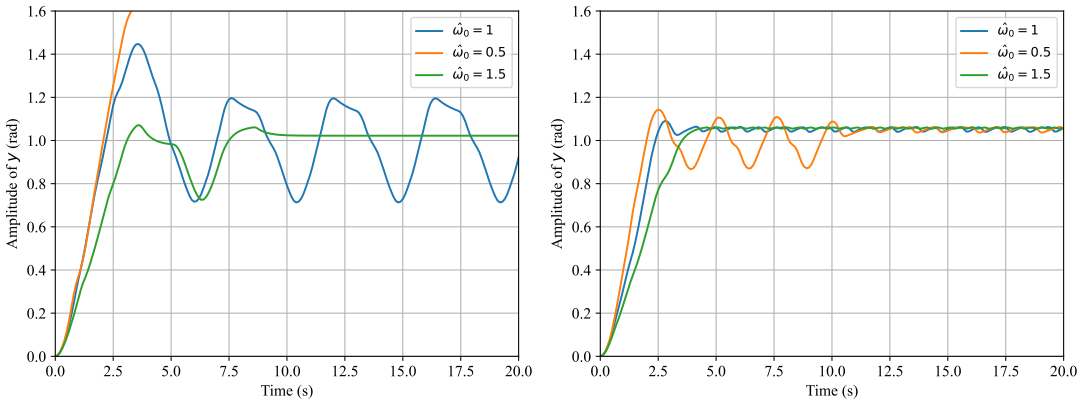
(**a**) Case 1 trained with 1000 episodes

(**b**) Case 2 trained with 10,000 episodes

(**c**) Case 3 trained with 20,000 episodes

(**d**) Case 4 trained with 40,000 episodes

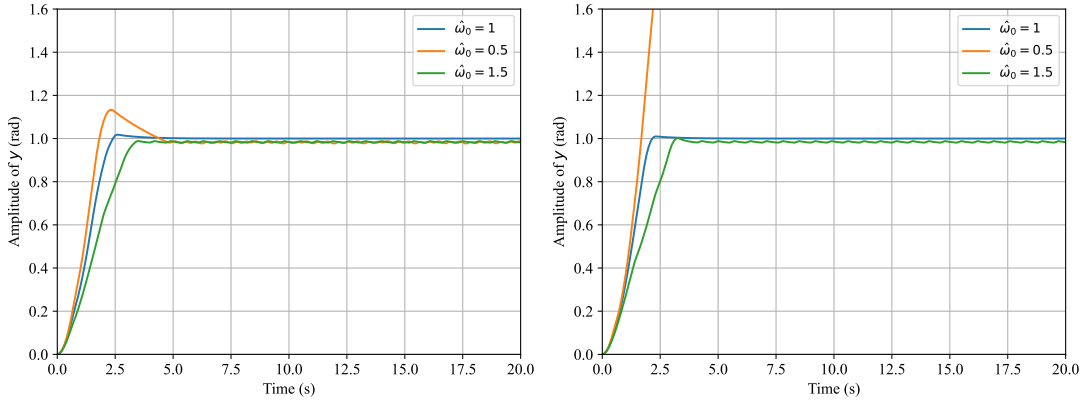
Figure 4. Cost-per-episodes of the system with Q-learning.



(**a**) Case 1 trained with 1000 episodes

(**b**) Case 2 trained with 10,000 episodes

Figure 5. *Cont.*



(**c**) Case 3 trained with 20,000 episodes (**d**) Case 4 trained with 40,000 episodes

Figure 5. Test of the trained optimal policies of Q-learning.

Considering the above tests, the third training with 20,000 episodes is chosen to make a comparison with the proposed EDR strategy with $\alpha = 4$. The results are shown in Figure 6.

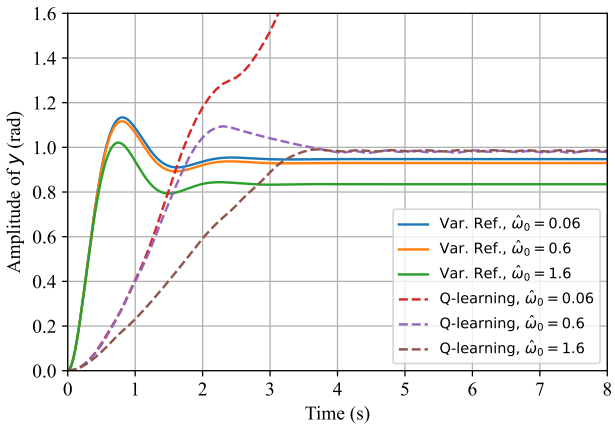


Figure 6. Comparison with Q-learning based control.

It can be observed that the Q-learning based control is able to regulate the system state x_1 around 1 for $\hat{\omega}_0 = \{0.6, 1.6\}$. When $\hat{\omega}_0 = 0.06$, the state evolves outside the state space and does not return back, failing the regulation. The reason is that, at $\hat{\omega}_0 = 0.06$, the system dynamics deviate significantly from the nominal training condition $\omega = 1$. The Q-learning policy, trained on a discretized state–action space around the nominal equilibrium, cannot generalize to this extreme parameter shift. As a result, the learned policy drives the pendulum outside the discretized state bounds $(-\pi/2, \pi/2)$, leading to uncontrolled divergence. As $\hat{\omega}_0$ increases from 0.6 to 1.6, the response slows down. On the other hand, the proposed EDR strategy can regulate all three systems to new equilibria. Although Q-learning achieves marginally smaller steady-state errors in the two successful cases of Q-learning-based control, this comes at a prohibitive cost: each policy requires extensive training and fails completely for $\hat{\omega}_0 = 0.06$. Moreover, to regulate the system to a new equilibrium different from $[1, 0]^\top$, retraining is generally necessary; otherwise, performance degrades significantly or instability may occur. In contrast, the EDR strategy requires no training, adapts instantly to new reference signals, and guarantees bounded tracking error across all tested model mismatches. This makes the EDR strategy more practical for real-world deployment, where reliability and efficiency are paramount.

4.4. Parameter Sensitivity Analysis

This sensitivity analysis demonstrates the skew rate parameter's influence by varying α while applying the controller (25) to a system with inconsistency $\hat{\omega}_0 \in [0.5, 1.5]$ and $\hat{\omega}_0 \in [0.1, 1.9]$. The results are depicted in Figure 7.

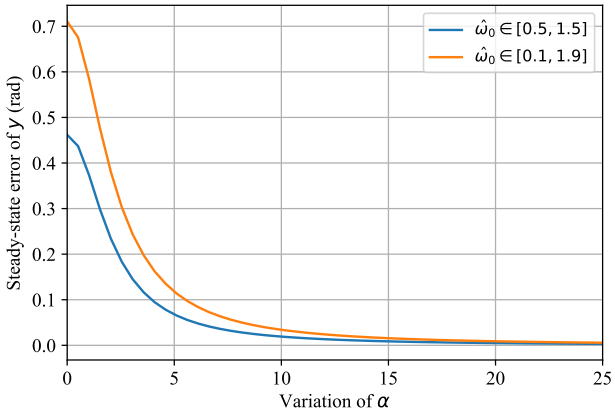


Figure 7. Impact of skew rate

The lines show the steady-state deviation x_1 for each value of α due to the system inconsistency. For any parameter $\alpha > 0$, the steady-state deviation is large for a large variation range $\hat{\omega}_0$, i.e., large system inconsistency. However, the induced deviation decreases monotonously with parameter α.

Note that we have implicitly used a "reduced-order" EDR due to the pendulum's structure. To show explicitly how the simulation adheres to the proposed EDR strategy, one may write the pendulum system as follows:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\hat{\omega}_0^2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ * \end{bmatrix}.$$

Then, the modification becomes the following:

$$\begin{bmatrix} u \\ * \end{bmatrix} = \begin{bmatrix} -3 & -4 \\ * & * \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 4 & 1 \\ * & * \end{bmatrix} \begin{bmatrix} r \\ 0 \end{bmatrix} + \Gamma \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} r \\ 0 \end{bmatrix} \right),$$

where Γ could be chosen as $\Gamma = -\alpha^2 I$. Theoretically, any nonzero α is admissible since $\Gamma = -\alpha^2 I$ is negative definite for all $\alpha \neq 0$. However, the control input takes the form $u = -(3 + \alpha^2)x_1 - 4x_2 + (4 + \alpha^2)r$, which implies that the control effort $\|u\|$ increases with α^2. On the other hand, the steady-state error decreases as α^2 increases. Therefore, α should be chosen large enough to meet the steady-state error specification with sufficient margin, while balancing the resulting increase in control effort.

4.5. An Example of MIMO LTI System

In the following, a numerical validation on a 2×2 multiple-input multiple-output (MIMO) LTI system is provided with parameter inconsistency. Consider the following:

$$\dot{x} = \hat{A}x + Bu, \qquad y = x, \tag{29}$$

where $x \in \mathbb{R}^2$, $u \in \mathbb{R}^2$, and $\hat{A}$ denotes the practical system matrix that may deviate from the nominal matrix A used in controller implementation. The reference is chosen as a nonzero setpoint $r \in \mathbb{R}^2$, so that equilibrium drift under parameter inconsistency can be observed. A model-based tracking controller with reference input can be written as follows:

$$u = -Kx + Nr, \tag{30}$$

where $K \in \mathbb{R}^{2\times 2}$ is a stabilizing feedback gain designed for the nominal model. The matrix N satisfies the following:

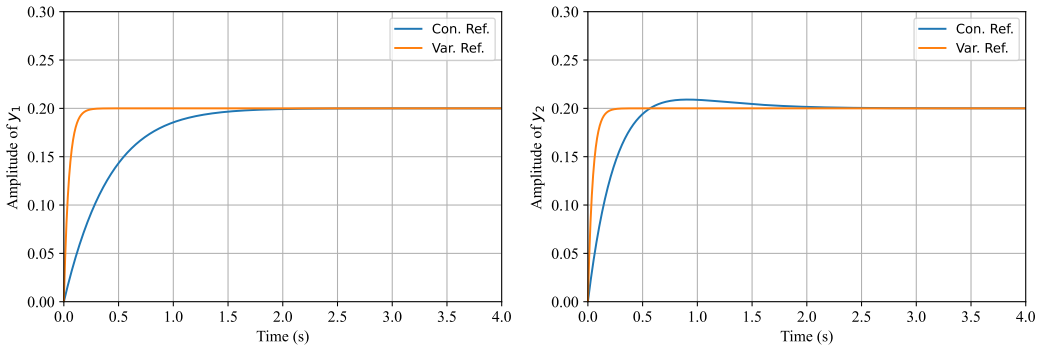
$$0 = (A - BK)r + BNr. \tag{31}$$

Following the proposed EDR strategy, we introduce an additional restriction term through the following:

$$u = -Kx + Nr + \Gamma(x - r), \tag{32}$$

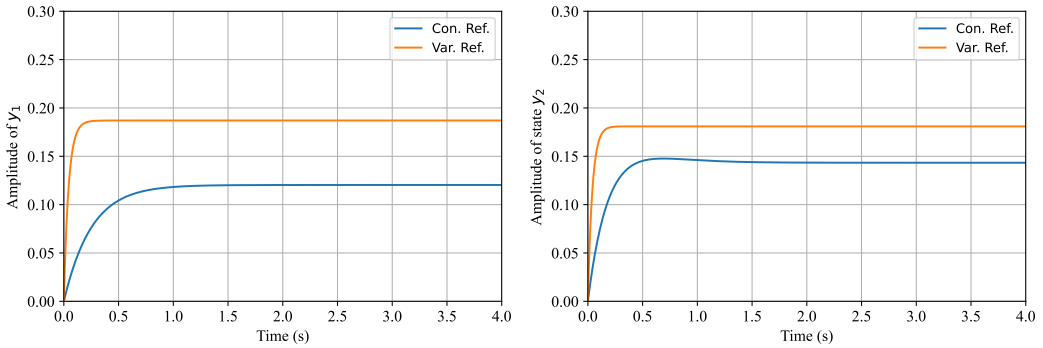
where Γ is designed as $\Gamma = -\alpha^2 I$.

In the simulation, we set the following: $A = \begin{bmatrix} -0.6 & 1.2 \\ -1.0 & -0.8 \end{bmatrix}$, $B = \begin{bmatrix} 1.0 & 0.2 \\ 0.1 & 1.0 \end{bmatrix}$, $r = \begin{bmatrix} 0.2 \\ -0.2 \end{bmatrix}$ and $K = \begin{bmatrix} 2.0 & 0.5 \\ 0.3 & 2.2 \end{bmatrix}$. In the simulation case in the absence of system inconsistency, the actual system matrix $\hat{A}$ is set to $\hat{A} = A$. In the simulation case under system inconsistency, the actual system matrix is set to $\hat{A} = \begin{bmatrix} -2.0 & 1.0 \\ -2.0 & -2.0 \end{bmatrix}$. The parameter α is set to $\alpha = 4$. The simulation results in the absence of system inconsistency and under system inconsistency are shown in Figure 8 and Figure 9, respectively.



(**a**) Amplitude of y_1 (**b**) Amplitute of y_2

Figure 8. Comparison of the MIMO LTI system response in the absence of system inconsistency between constant and variant reference input.



(**a**) Amplitude of y_1 (**b**) Amplitute of y_2

Figure 9. Comparison of system response under system inconsistency between constant and variant reference input.

Figure 8 shows that, under the nominal model, both the controller (30) and the controller (32) regulate the state to the desired setpoint r. In contrast, Figure 9 demonstrates that, when $\hat{A} \neq A$, the controller (30) with fixed-reference tracking exhibits a clear equilibrium drift and a non-negligible steady-state error, while the controller (32) reduces the steady-state deviation as expected.

5. Conclusions

This paper establishes equilibrium drift restriction as a principled strategy for mitigating steady-state deviations induced by system parameter inconsistency. The approach fundamentally rethinks the reference input method in control systems by introducing the intersection point-of-view of equilibrium. Both geometric interpretations and theoretical analysis have demonstrated how this modification constrains parameter-induced deviations. Explicit deviation bounds under system inconsistency have also been analyzed. Validation on the pendulum system confirms the strategy's dual capability: preserving nominal performance while enhancing robustness against substantial parameter variations. The comparative study with Q-learning further highlights its operational advantage in setpoint adaptability. Parameter sensitivity analysis has additionally quantified how tuning the restriction strength reduces steady-state error.

6. Patents

The following are the patent results from the work reported in this manuscript: Fangyuan Li. A Method for Controlling setpoint Tracking Accuracy Based on Variable Inputs. Chinese Patent ZL202411086897.4, granted 30 September 2025.

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Informed Consent Statement: Not applicable.

Data Availability Statement: The source code and data used in this study are openly available in the GitHub repository at https://github.com/fylimas/EDR, accessed on 13 November 2025.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

EDR	Equilibrium drift restriction strategy
LTI	Linear time-invariant
MIMO	Multiple-input multiple-output

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